

Assignment on Java Collection Framework

1. Customer Service Simulation

Simulate a customer service queue where customers are served in the order they arrive (FIFO).

- Use `Queue<String>` to add customer names.
- Provide functions to add, serve, and peek at the next customer.

2. Print Job Scheduler

Model a print queue system using `Queue<String>` where each string represents a document.

- Add 5 documents.
- Process (remove) 2 documents.
- Display the remaining queue.

3. Palindrome Checker Using Deque

Use `Deque<Character>` to check if a word is a palindrome.

4. Circular Queue Logic (custom implementation)

Implement your own circular queue using an array of size 5. Provide `enqueue`, `dequeue`, `peek`, and `isFull` methods.

5. Unique Words from a Sentence

Input a sentence and extract all unique words using a `HashSet<String>`.

6. Duplicate Remover

Given a list of integers, remove all duplicates using a `Set<Integer>`.

7. Common Elements Between Two Sets

Input two sets of integers. Find their:

- Intersection
- Union
- Difference

8. Employee Skill Tracker

Given employees and their skill sets:

Find:

- Employees with a specific skill
- Common skills between two employees

9. Word Frequency Counter

Input a paragraph and count the frequency of each word using `Map<String, Integer>`.

10. Student Marks Record

Use `Map<String, Integer>` to store student names and marks.

- Add at least 5 students.
- Find the student with the highest and lowest marks.

11. Character Count

Given a string, count the occurrences of each character using a `HashMap<Character, Integer>`.

12. Phone Directory

Implement a phone book using `Map<String, String>` where key = name, value = phone number.

Provide:

- Add contact
- Search contact
- Remove contact

BONUS Question (Optional)

Task Scheduler

- Maintain a `Map<String, Queue<String>>` where each key is a day (e.g., "Monday") and the value is a queue of tasks.
- Allow adding, viewing, and removing tasks for each day.

Library Book Management

- Use `Map<String, Set<String>>` where the key is a genre (e.g., "Fiction") and the value is a set of book titles.
- Add, remove, and search books by genre.

Voting System

- Use a `Map<String, Integer>` to store candidate names and their vote count.
 - Add votes and find the winner.
- 1.